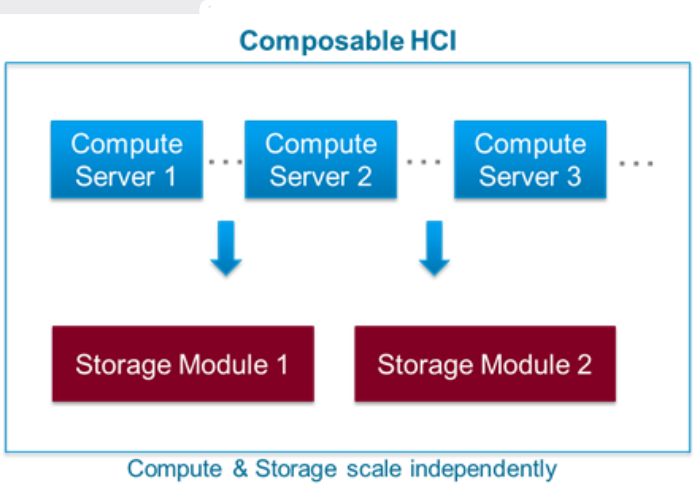
Process: design challenges

Types of infrastructure (1/2)

1 Composabl

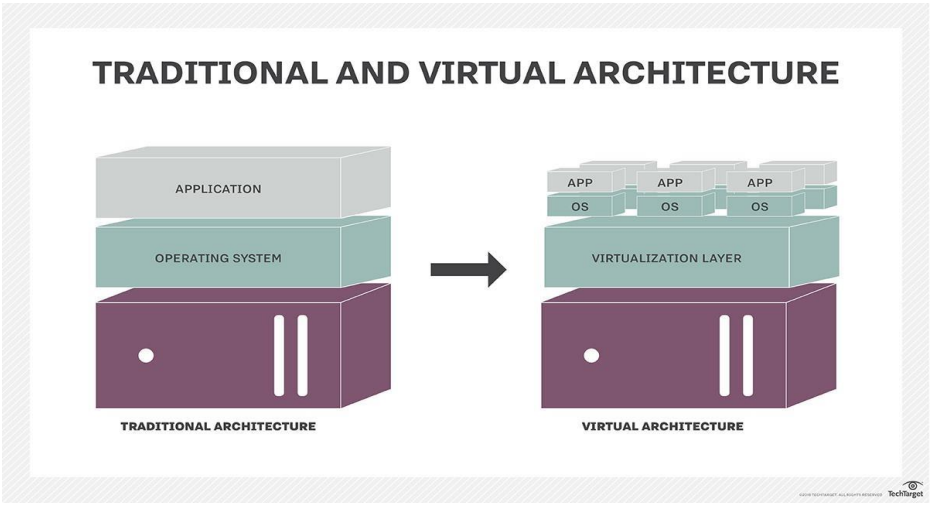


Source:
https://blogs.vmware.com/virtualblocks/2017/05/04/hpe-synergy-first-composable-infrastructure-now-certified-vsan/

2 Cloud



3 Infrastructure as a Service (IaaS)

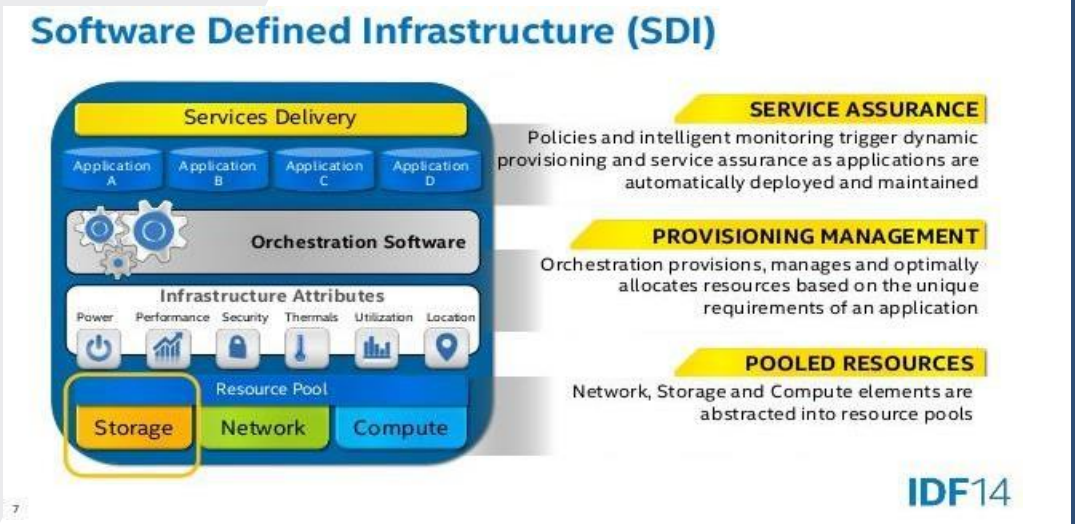


Source:
https://www.researchgate.net/publication/318912754_How_Software_is_Transforming_the_US_Economy

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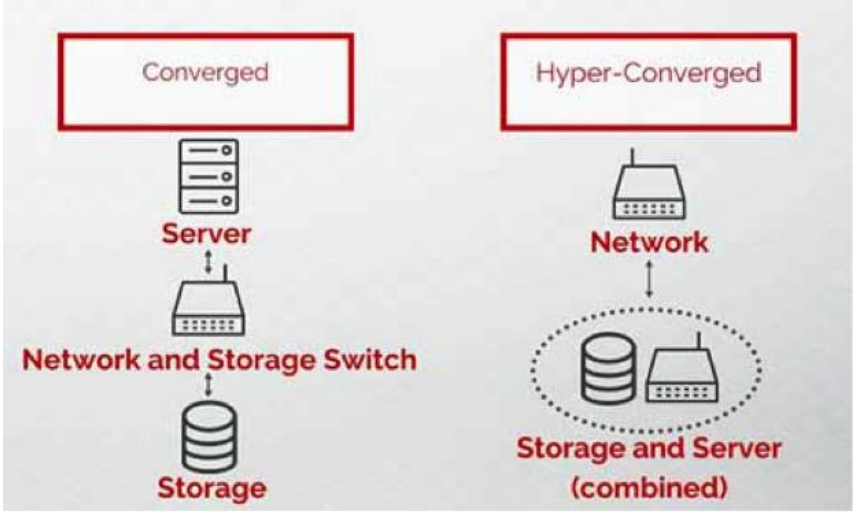
Types of infrastructure (2/2)

4 **Software-defined**



Source: https://www.researchgate.net/publication/318912754_How_Software_is_Transforming_the_US_Economy

5 **Converged/Hyperconverge**



Source: https://www.nutanix.com/theforecastbynutanix/technology/difference-between-converged-and-hyperconverged-infrastructure

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Infrastructure as Code (IaC)



Idempotent

Mathematical operation that always returns same result, no matter how often the input is entered (Source: Dictionary.com) Example: deployment command always sets target into same configuration, regardless of starting state.

The DevOps way to do it

- Hardware and software convert to virtual.
- Machine-readable files test and run code.
- Evolved from hardware virtualization.
- Maintains mirrored environments.
- Prefers declarative delivery.

Benefits

- improved system performance
- speed
- change management
- scalable
- customer satisfaction.

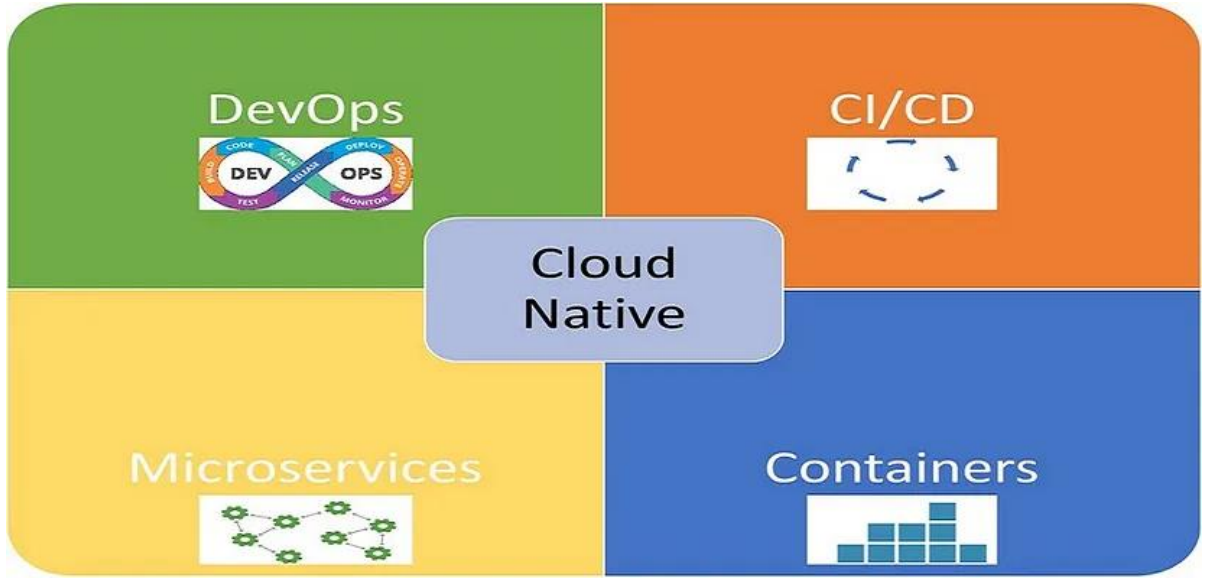
Challenges

- New skills are required.
- Requires periodic audit.
- Needs efficient planning.
- Collaboration is key.

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Cloud and cloud-native (1/2)

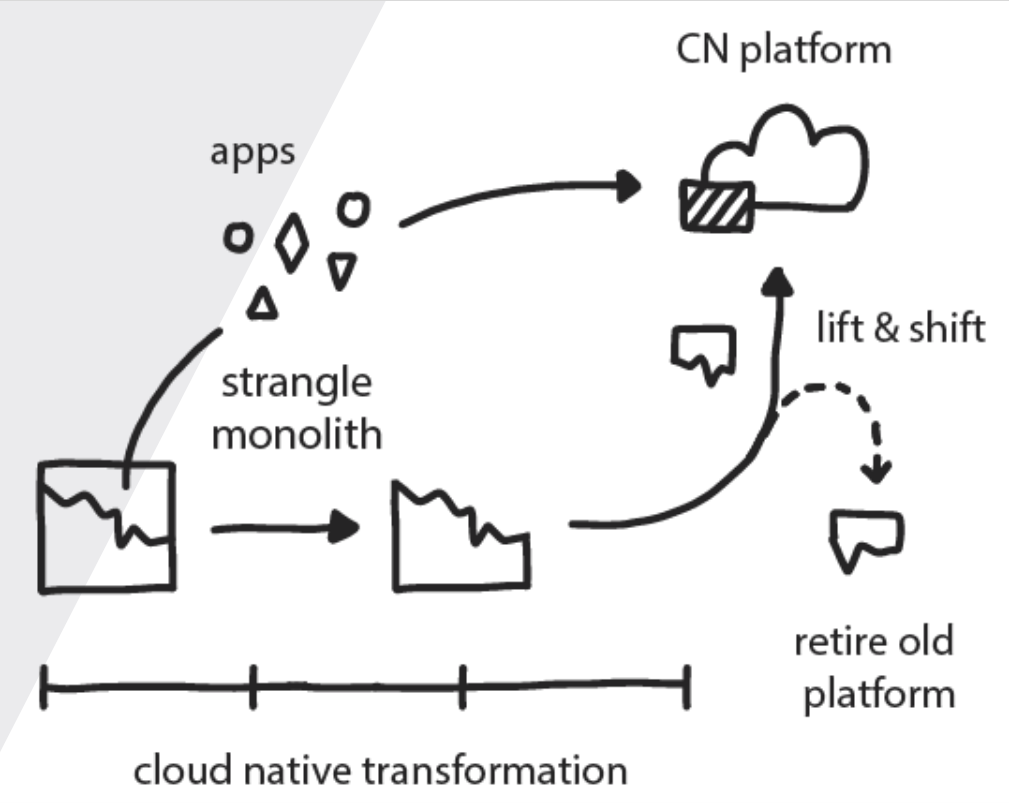
- Partial lift and shift; some apps, part of the time.
- Choose fully provisioned services.
- May need to learn to be cloud-migrant.
- Some services, some legacy
  - Establish a value stream.
  - Be a smart shopper.



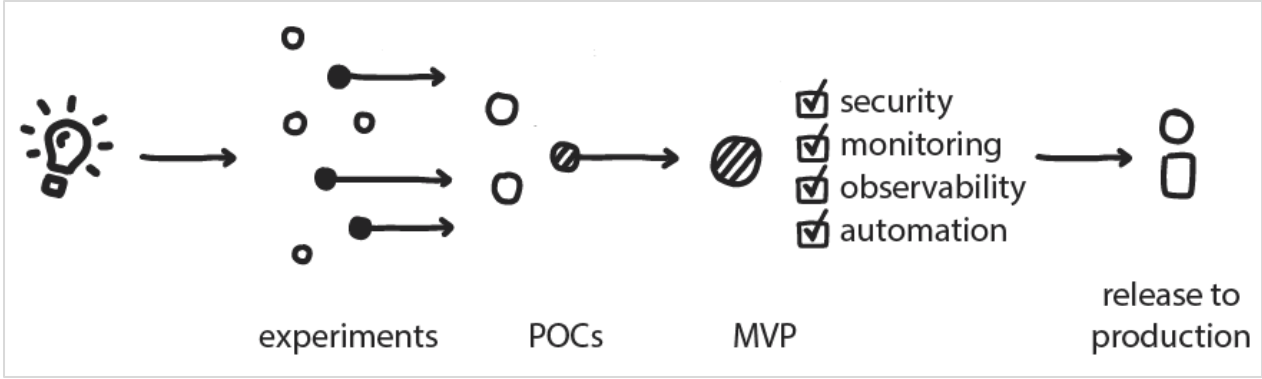
Source: https://satyanprakash.medium.com/the-road-to-cloud-native-transformation-goes-through-devops-461f3a652ead

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Cloud and cloud-native (2/2)



Source: https://www.cnpatterns.org/development-design/strangle-monolithic-application

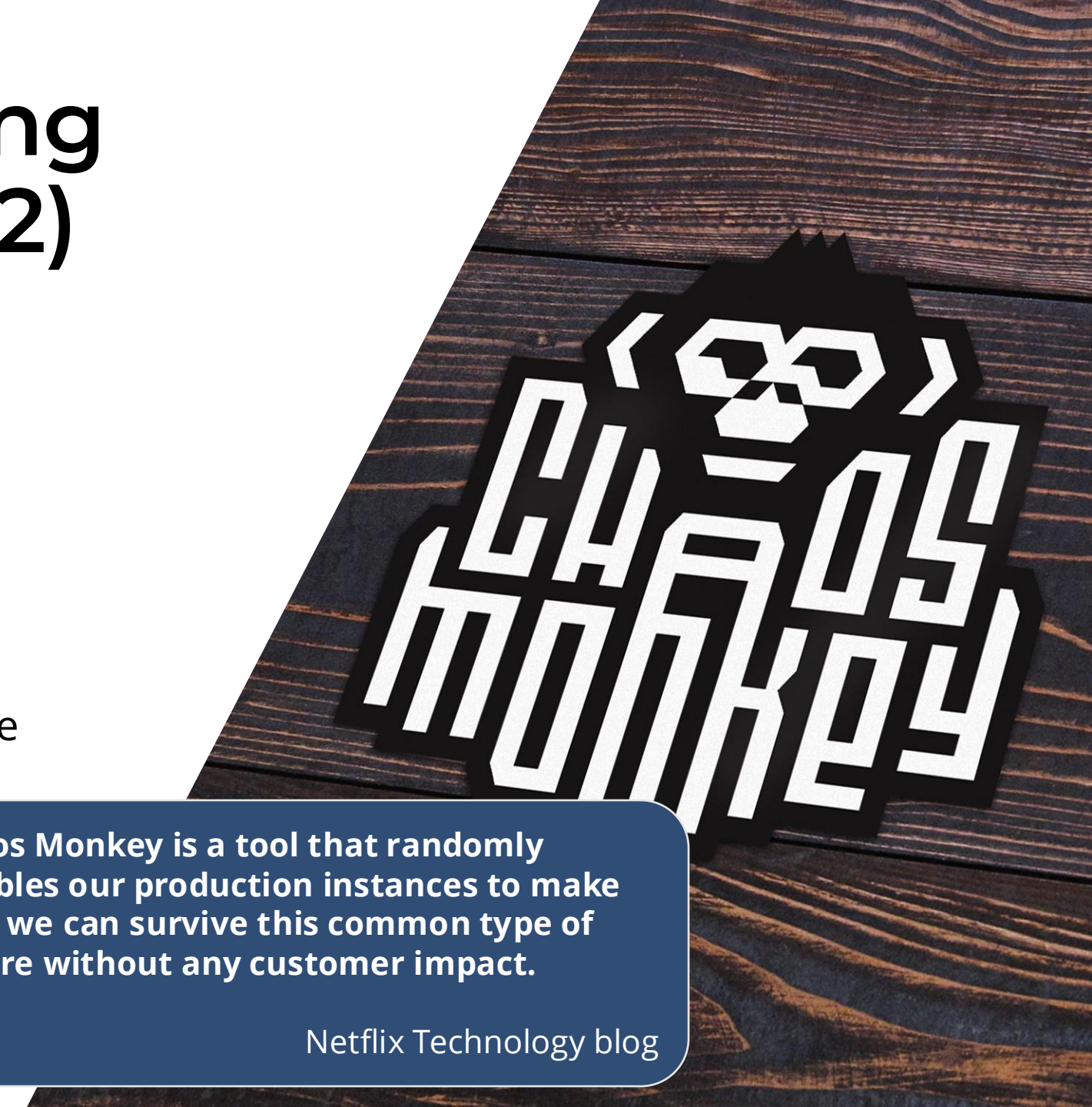


Source: https://www.cnpatterns.org/infrastructure-cloud/full-production-readiness

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Chaos engineering infrastructure (1/2)

- Is similar to resilience engineering.
- Disciplined approach to identify vulnerabilities in production
  - Checks reliability, stability, and capability.
  - Improves resilience.
  - Tends to be proactive.
- Testing is planned and limited in scope
  - Chaos is wide, unpredicted experiments to generate new knowledge.
- Discover what if scenarios.



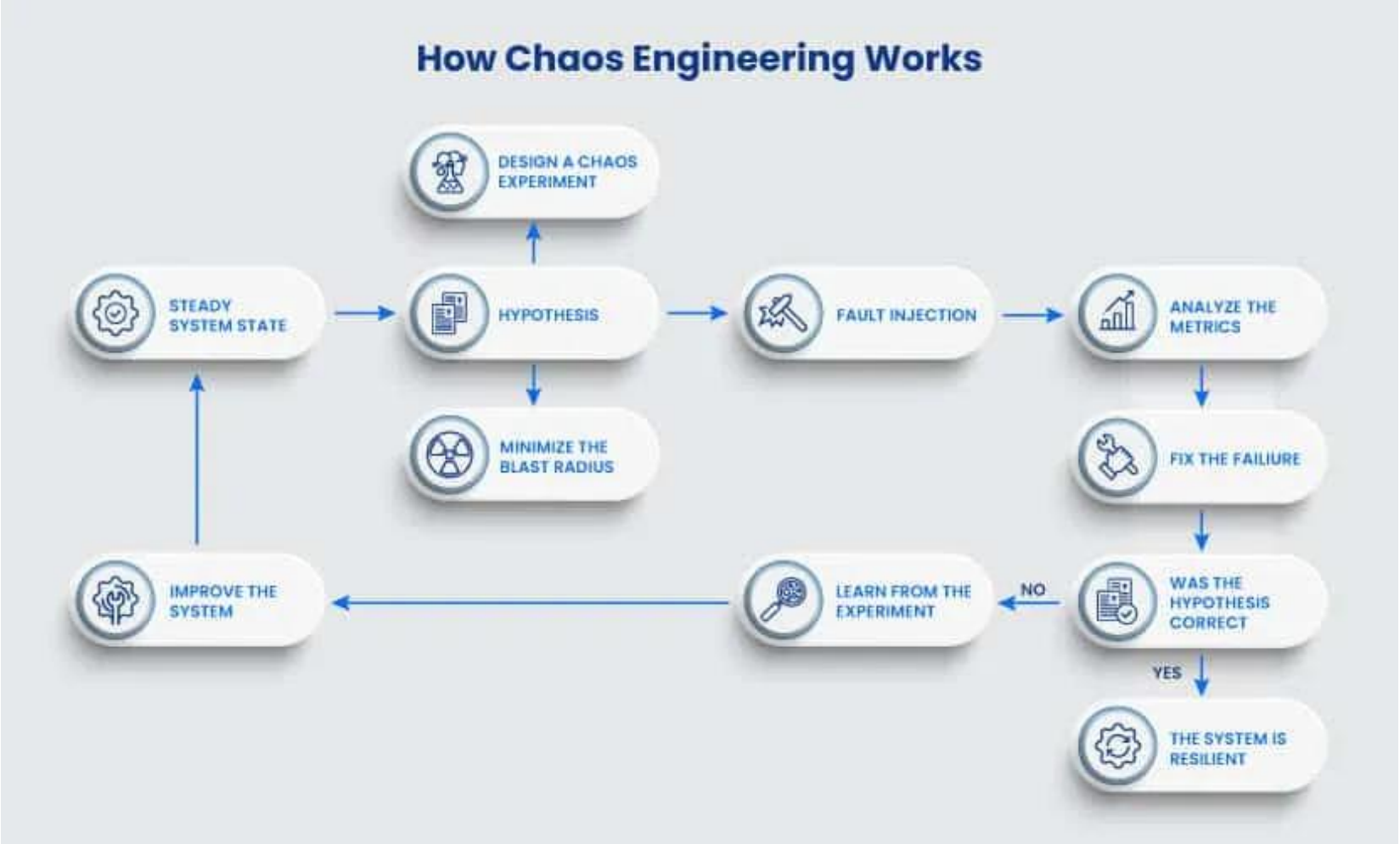
Chaos Monkey is a tool that randomly disables our production instances to make sure we can survive this common type of failure without any customer impact.

Netflix Technology blog

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Chaos Engineering infrastructure (2/2)

The bigger the infrastructure , the more chaos.

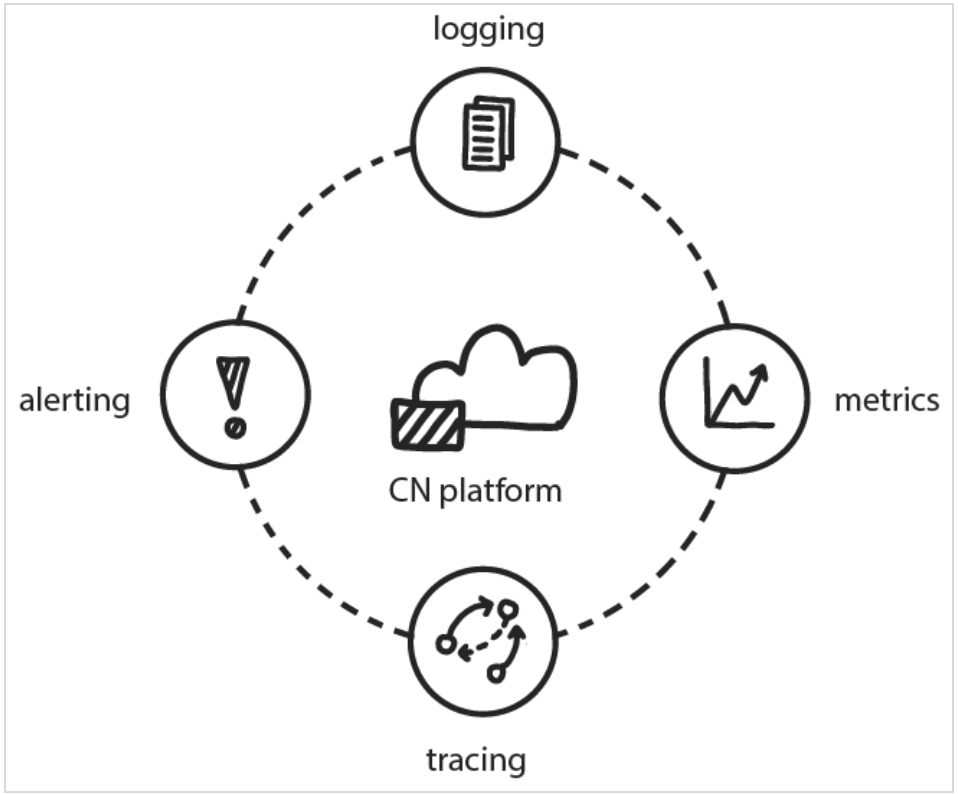


Source: https://medium.com/devopscurry/how-chaos-engineering-plays-a-vital-role-in-devops-success-68ba35307f1b

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Securing your infrastructure (1/2)

- **Firewall/proxy:** filters traffic with policies and rules.
- **Authenticate with username/password.**
- **Delegated authentication**
  - two-legged authentication
  - three-legged authentication using Custom App and Identity Cloud Service (IDCS).
- **Two-factor (or more) Authentication (2FA, MFA)**
  - something you have (Fob)
  - something you know (password or PIN)
  - something you are (biometric).
- **Certificate-based authentication**
  - Secures apps system to system without user interference.

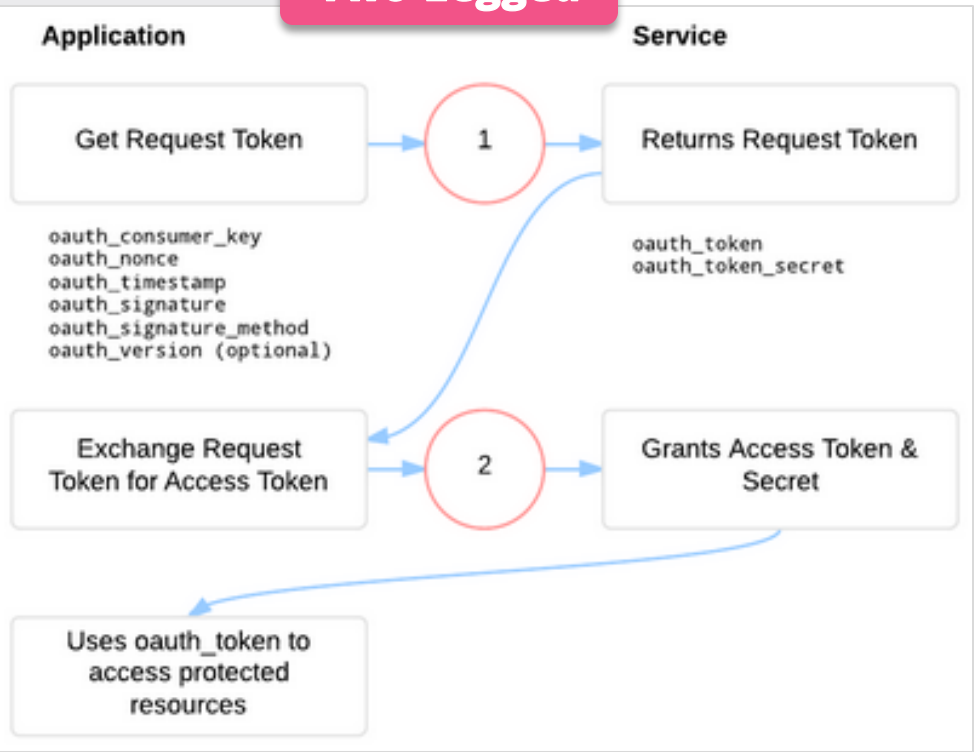


Source: https://www.cnpatterns.org/infrastructure-cloud/observability

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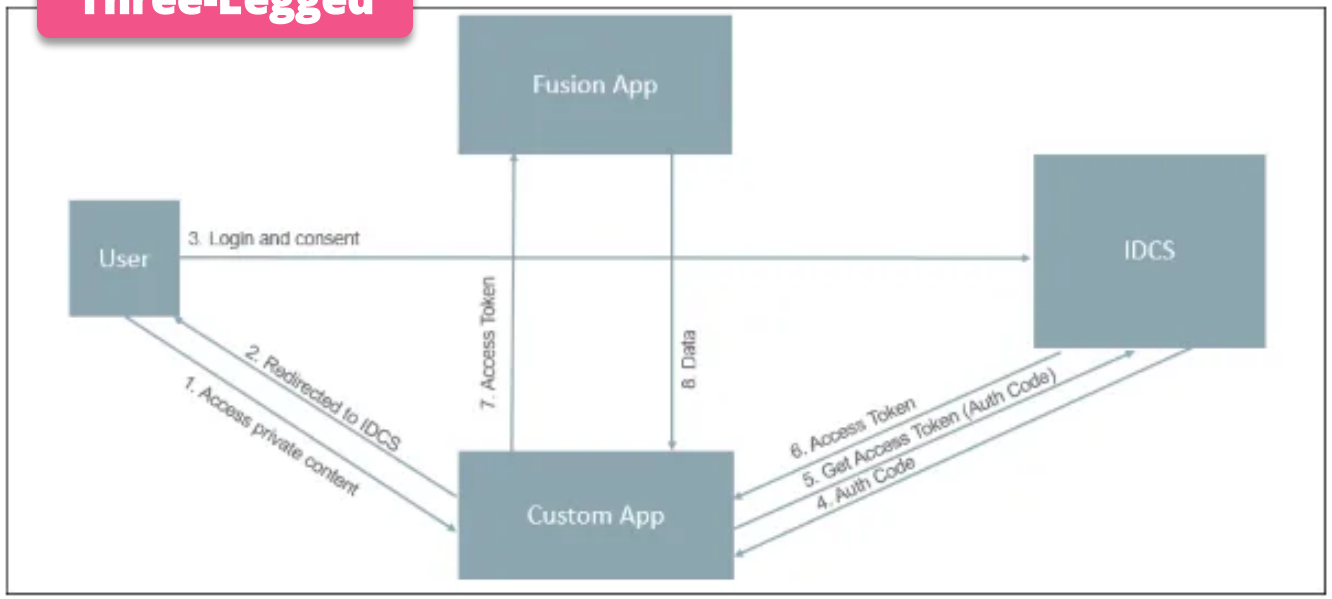
Securing your infrastructure (2/2)

Two-Legged



Source: https://github.com/enisinanaj/carinfo-importer?tab=readme-ov-file

Three-Legged



Source: https://medium.com/oracledevs/3-legged-oauth-flow-to-invoke-fusion-apps-rest-endpoints-5fc6b0b7b059

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